Adaptability — Implementation

What Is Built, What Has Run, and the One Place They Do Not Meet

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DRAFT — draft-2026-09-02.1 r6 (2 September 2026). Circulated for comment, not as a finished result. It is a companion to *Adaptability* (referred to below as the adaptability paper), to *The Law of Graphic Equalisation — Theory* (Graphic Equalisation) and to *Assessing an Account of This Kind* (the assessment companion), and inherits their definitions, epistemic boundary, and level-of-claims; those short names are the only forms used below. The adaptability paper states the theory; this paper reports the implementation and its principal gap. Sections marked as falsifiability commitments are the intended points of attack.

Abstract

The operation the adaptability paper describes is built, has been measured on the agent's own corpus, and has been corrected against itself, that correction being the principal result reported here. It is also, in one specific and consequential way, in the wrong place. *The algebra is native; the faculty that uses it is not.*

The standing of what is reported here should be said once and then relied on. Every figure below comes from this programme's own pre-registered implementation record and from the working tree that record describes, neither of which is circulated with these papers, and each is therefore a claim a reader may decline to credit on exactly the terms Graphic Equalisation's Section 1 sets out. What is offered against that is exactness rather than authority: *the counts in this paper are generated from the tree rather than transcribed into it*, at commit `f327b151730c` read on 2 September 2026, by a script kept beside the paper and re-run at every revision, so a reader who does hold the tree can re-run them and a reader who does not can at least see what was counted.

1 What is built

The transform algebra is native to the engine. Anti-unification retaining its witness, substitution, unification, structural similarity and a coverage predicate are all term-level operations, evaluable inside the rule language rather than beside it. The shape ladder is built on them: decoding against a held codebook, transformation within the held basis, inference of a dimension, and minting.

The ladder runs as rules. It is a rule fixpoint rather than a loop in the host language — a residual mints a quantum, a codebook epoch re-arms decoding, and the re-decode either recognises or does not. That much is graph-resident and gated.

2 What has run

Remark 2.1 (Transfer by parallelogram completion). The strongest demonstration takes four of the agent's own rules arranged as a 2×2 over two independent dimensions. *Parallelogram* is borrowed from the embedding-arithmetic literature and the adaptability paper's Section 14 sets out the lineage, along with the reason that literature's known critique does not carry over to an exact symbolic operation. Anti-unification over two corners derives the transform; substitution applies it to the third; the result is a rule *absent from every distinct recogniser the corpus held* — 2,851 of them when the transfer was measured, on 12 August 2026 — which, installed, fires correctly on the missing cell and on no other. *That one is verified behaviourally.*

318 such families exist in the corpus, and 200 of 200 sampled transfers are algebraically lossless, well-defined and novel. *Algebraically is the whole of what that second figure says.* One transfer was installed and run; the other 199 establish that the operation is well-defined and that its output is not already held, and not that the derived rule fires. Reading 200 of 200 as 200 installations would be reading a stronger result than was taken. *No new term variants, no graph mirror and no engine change were required* — the capability was reachable with what already existed.

Remark 2.2 (The concept-grain faculty, and its own correction). At the concept grain the faculty has run against the agent's largest real bridge, 142 concepts against 137, and reported 8,575 analogies.

Of those, 8,253 *were an artefact*: a keyword matching *inside* a longer word, which accounted for 39.6% of every vocabulary hit in the booted graph. After token-anchoring and a requirement of structural support the count fell to 321, of which 78 are corroborated by a witness shared across pairs.

That is reported here as the faculty's principal measured result, not as an embarrassment adjacent to one. A ninety-six per cent retraction is what the correction was worth. *What this paper stands behind is the withdrawal*, which rests on a stated mechanical defect in the matcher and applies to every one of the 8,253. It does not stand equally behind the survivors: nobody has read the 321 and confirmed each, so 321 is an upper bound with 78 corroborated inside it, and a reading of them could move it down again.

3 The gap

The two halves above do not meet, and the seam is exactly locatable.

Anti-unification with witness retained appears 7 times in the faculty's host-language module and 0 times across the 3,843 rules carried by the 805 seed manifests. The faculty is 1,241 lines of host code, and it *does* call the native algebra — so this is not a capability that was built and left unused. It has run, it has been measured, and its output survives correction.

What it is not is graph-resident. The transform algebra is reachable from the host language and not from a rule.

Remark 3.1 (Why that matters here more than anywhere else). This programme's boundary on self-repair is sharp, and it is a standing rule of the implementation record rather than a finding of this paper: *the agent can repair and extend its cognition exactly where that cognition is graph-native*, because what it can vary, inspect and mint against is graph data. A faculty in the host language can be used and cannot be improved by the thing using it.

So of all the faculties that could have been left outside, *this is the worst one* — because it is the adaptation mechanism. The system's ability to acquire competences it was not given is itself a competence it cannot acquire. That is not a contradiction, but it is a ceiling, and it is the ceiling the adaptability paper's ladder (its Section 3) was built to remove.

The measured retraction above sharpens it. A ninety-six per cent artefact rate was found by an audit performed *on* the faculty from outside. A graph-resident faculty is the kind of thing

this substrate has repeatedly been made to notice about itself — it files gaps, retires its own duplicates, and surfaces what it cannot answer. *The faculty most in need of that treatment is the one that cannot receive it.*

4 What the gap is not

It is not that the algebra is unreachable in principle. Neighbouring operations from the same family are already invoked by rules, and the census is exact rather than impressionistic — a match on the term as a rule actually carries it, not on the term's name appearing somewhere in a file. The coverage predicate is invoked 12 times, across 3 seed manifests, and its open variant 6 times. The *witness-discarding* anti-unification — the classical least general generalisation, which returns the generalisation and throws away the substitutions that recover its inputs — is invoked 11 times, across 3 manifests. Structural similarity is invoked 2 times.

Unification is invoked 0 *times, and substitution* 0. The substitution operation's name does occur as text in 3 manifests, none of which invoke it — which is what a looser search returns, and is why this census is taken on the term rather than on the text.

That is a sharper finding than a bare absence, and it points the same way. The vocabulary the rule layer actually holds is the classical one: cover a case, and generalise two structures while discarding what distinguished them. The two operations that carry a transform — retaining the witness, and re-applying it — are exactly the two that no rule invokes. The adaptability paper's Section 14 states the consequence in its own terms: an lgg that throws away its witness cannot adapt anything.

It is not that the demonstration is unsound. The parallelogram result stands on its own terms, and its numbers are reported above with the caveat they deserve: it was demonstrated in analysis scripts rather than as agent behaviour, so it establishes that the corpus *contains* 318 derivable rules, not that the agent derived them.

And it is not a large amount of work in prospect. The absence of new term variants in the demonstration is the relevant measurement: what would be required is a rule surface over an algebra that already evaluates, which is the same move this programme has made before and reported.

5 A second instance of the same seam

The pattern of Section 3 is not confined to analogy, and the clearest other case is the one most directly about acquisition: reading a scientific paper and minting from what it says.

The path is built and is the ladder pointed at literature. An adapter presents a retrieved document as a typed graph; the agent navigates a preprint archive, retrieves a paper, and grounds it. Grounding the abstract yields the paper's *subject* — and the record's own note on that step is a negative rather than a success: *grounding to primitive was false*, so the agent knew what the paper was about and not what it claimed. The body pass exists to close exactly that gap: the claim-bearing environments are parsed, and each claim is grounded through the construction grammar — a term as a genus plus its differentia, recursively toward primitives already held — carrying a *depth* alongside the result, so that grounded-to-topic and grounded-to-primitive are distinguishable outcomes rather than one verdict.

And where a claim rests on a construction the agent does not hold, that is detected and queued to acquire. That is rung four with the residual named: not a failure to read, but a located absence with a next action attached.

Remark 5.1 (Its status, stated exactly)*. Two things are built here, only one of them is gated, and they are reported apart because reporting them together would credit the second with the first's standing.*

The grounding faculty the path stands on — taking a definition in prose and descending it toward primitives already held — is gated. Five tests cover it, two of them confirming that its seed is load-bearing and installs only through the loader, so the faculty is not an artefact of a script arranging its own inputs. *What those tests exercise is the faculty on prose fed to it.* They do not exercise the retrieval, the abstract-and-body split, or the depth distinction.

Those are built and are not gated. The retrieval path and the body pass exist and their arc is recorded in their own source; no test drives them. *And no results document exists for either half.* There is no figure for how many claims ground to primitive against topic, none for how often an un-held construction is correctly detected, none for what happens to the queue. So the status is a gated faculty, an ungated path built on it, and no measurement of the arc end to end — and this paper reports it at that status rather than at the one the machinery invites.

And the seam is the same one. The grounding faculty is host-resident, with a single seed referencing it. So the two capabilities this programme has for acquiring what it was not given — transfer between held rules, and reading claims out of a document — *are both real, both only partly gated, and both sit outside the boundary within which the system can repair itself.*

6 The contrast the record supplies

One further measurement belongs here, because it is uncomfortable in the right direction.

Program synthesis in this substrate is *graph-native* in the way analogy is not: its enumerator is a term, and the decisions — propose, evaluate, accept — are seed rules. And its clearest result about held content is a null. Running the loop with the acquired content and running it stripped produced 54 of 257 in both arms, and not merely as equal totals: *the two arms returned byte-identical result sets*, the same tasks solved and the same tasks missed.

So the record contains both of these at once. *The faculty that is graph-native produced a clean null about the value of what it held; the faculty that is not produced the strongest positive result in the corpus, in analysis scripts.* Neither observation rescues the other, and the pair is reported together because taking either alone would misrepresent the record — the first is not a reason to doubt the second, and the second is not a reason to discount the first.

The reading that survives both is narrow, and the adaptability paper's Remark 6.1 states its general form: capability has plural sources selected by regime, so which faculty happens to be graph-resident says nothing about which one carries a given domain. *Being graph-resident is what makes a faculty improvable by the system carrying it, and it is not what makes a faculty work.* The gap in Section 3 is a ceiling on self-repair, and this section is the reminder that removing it would not by itself buy a capability.

7 Falsifiability commitments

Each commitment below is marked [SHAPE] or [DETAIL], in the sense of Graphic Equalisation's Section 19. A [DETAIL] failing is a revision: the account survives with that mechanism replaced. A [SHAPE] failing costs *range* rather than a mechanism, because what fails is something the substrate is claimed to require or to permit — so the account does not hold where it claimed to, which is the more serious of the two and is still a boundary rather than an annihilation.

1. [DETAIL] **The two transform-carrying operations are absent from the rule layer.** Fails on the exhibition of any seed rule invoking witness-retaining anti-unification *or* substitution. Both counts are 0 across the 3,843 rules of the 805 manifests, taken at commit `f327b151730c` on 2 September 2026, and are a command rather than a judgement.

2. [SHAPE] **Self-repair is bounded by graph-nativeness.** Fails if the agent can be shown to improve a host-language faculty by its own operation rather than by an author's — which would remove the ceiling this paper reports and much of its point.

3. [DETAIL] **The 318 families are algebraically derivable and novel.** Fails if a sampled transfer is found to be lossy, ill-defined, or to reproduce a recogniser already in the corpus.

4. [DETAIL] **A transfer that passes the algebra fires.** Fails if a derived rule that is lossless, well-defined and novel, once installed, does not fire on its missing cell or fires elsewhere. One case has been run; this commits the other 317.

5. [DETAIL] **The corrected analogy count is the defensible one.** Fails if token-anchoring and the structural-support requirement can be shown to discard genuine correspondences rather than artefacts, which would make 321 an over-correction rather than a correction — or, in the other direction, if reading the 321 finds a material fraction of them spurious, which would make 321 an upper bound that has not yet been brought down.

6. [DETAIL] **The body pass distinguishes topic from primitive.** Fails if, on a paper whose claims rest only on constructions the agent already holds, the body pass grounds no claim to primitive — in which case the depth carried alongside the result is recording a distinction the machinery does not in fact draw.

Scope note (not a falsifier). The parallelogram result is about the corpus, not the agent. No claim is made that the agent derived those rules, and a reader who extracts one has found an error rather than a result.